15.6 Senescence

Plant life comprises some sequential events, *viz*: germination, juvenile stage, maturation, old age and death. Old age is called **senescence** in plants. Senescence refers to all collective, progressive and deteriorative processes which ultimately lead to complete loss of organization and function. Unlike animals, plants continuously form new organs and older organs undergo a highly regulated senescence program to maximize nutrient export.

The branch of botany which deals with ageing, abscission and senescence is called **Phytogerontology**

1. Types of Senescence

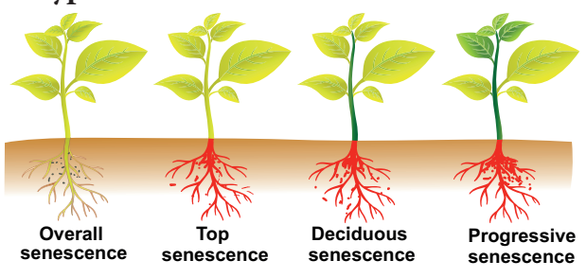


Figure 15.13: Different types of senescence in plants

Leopold (1961) has recognised four types of senescence:

i. Overall senescence
ii. Top senescence
iii. Deciduous senescence
iv. Progressive senescence

i. **Overall senescence**: This kind of senescence occurs in annual plants when entire plant gets affected and dies. Example: Wheat and Soybean. It also occurs in few perennials also. Example: *Agave* and Bamboo.

ii. **Top senescence**: It occurs in aerial parts of plants. It is common in perennials, underground and root system remains viable. Example: Banana and *Gladiolus*.

iii. **Deciduous senescence**: It is common in deciduous plants and occurs only in leaves of plants, bulk of the stem and root system remains alive. Example: Elm and Maple.

iv. **Progressive senescence**: This kind of senescence is gradual. First it occurs in old leaves followed by new leaves then stem and finally root system. It is common in annuals (Figure 15.13).

2. Physiology of Senescence

- Cells undergo changes in structure.
- Vacuole of the cell acts as lysosome and secretes hydrolytic enzymes.
- The starch content is decreased in the cells.
- Photosynthesis is reduced due to loss of chlorophyll accompanied by synthesis and accumulation of anthocyanin pigments, therefore the leaf becomes red.
- There is a marked decrease in protein content in the senescing organ.
- RNA content of the leaf particularly rRNA level is decreased in the cells due to increased activity of the enzyme RNAase.
- DNA molecules in senescencing leaves degenerate by the increased activity of enzyme DNAase.

3. Factors affecting Senescence:

- ABA and ethylene accelerate senescence while auxin and cytokinin retard senescence.
- Nitrogen deficiency increases senescence whereas nitrogen supply retards senescence.
- High temperature accelerates senescence but low temperature retards senescence.
- Senescence is rapid in dark than in light.
- Water stress leads to accumulation of ABA leading to senescence.

4. Programmed cell death (PCD)

Senescence is controlled by plants own genetic programme and death of the plant or plant part consequent to senescence is called **Programmed Cell Death**. In short senescence of an individual cell is called **PCD**. The proteolytic enzymes involving PCD in plants are **phytaspases** and in animals are **caspases**. The nutrients and other substrates from senescing cells and tissues are remobilized and reallocated to other parts of the plant that survives. The protoplasts of developing xylem vessels and tracheids die and disappear at maturity to make them functionally efficient

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